

Part 2: Technical Implementation

AI Crypto Hedge Fund MVP: data, models, backtests, portfolios

2.1

Baseline

2.2

Models + agents

2.3

Static portfolio

2.4

**Dynamic
rebalance**

2.5

120 pairs

One reproducible notebook and modular Python package covering data preparation, model validation, strategy backtesting, result visualization, and explanation.

1. Requirements Coverage

Implementation is structured exactly by the five assignment levels

Level	Requirement	Committed evidence
2.1	Single-crypto baseline strategy	scripts/run_baseline_strategy.py, baseline_metrics.json
2.2	Econometric, ML, and AI-agent single-pair comparison	run_single_asset_models.py, model and agent modules
2.3	5-7 coin static portfolio on historical data	run_static_portfolio.py, six-asset universe
2.4	Dynamic rebalancing for small portfolio	run_dynamic_rebalancing.py, rebalance events
2.5	100+ pair dynamic portfolio expansion	run_large_universe.py, 120-pair universe

Reproducible

uv, locked dependencies, smoke data, final notebook

Modular

src/ai_crypto_hedge_fund/ contains reusable code

Validated

Chronological train/test split, no shuffling, out-of-sample metrics

2. Data Preparation and Validation

All experiments share the same timestamped minute-bar data contract

Data snapshots

Committed sample data supports offline smoke tests. The full 1-minute 120-pair processed bundle is documented by manifest, checksums, and external URL.

Model validation

Time series split: 70% train, 30% test. No random shuffle. Signals are shifted one period before returns are applied.

Universe construction

Small universe: BTC, ETH, BNB, SOL, XRP, ADA. Large universe: 120 Binance USDT spot pairs selected by coverage and liquidity filters.

Reproducibility checks

`uv run pytest`, notebook execution with `DATA_MODE=sample`, JSON validation for manifest, notebook, and all metric reports.

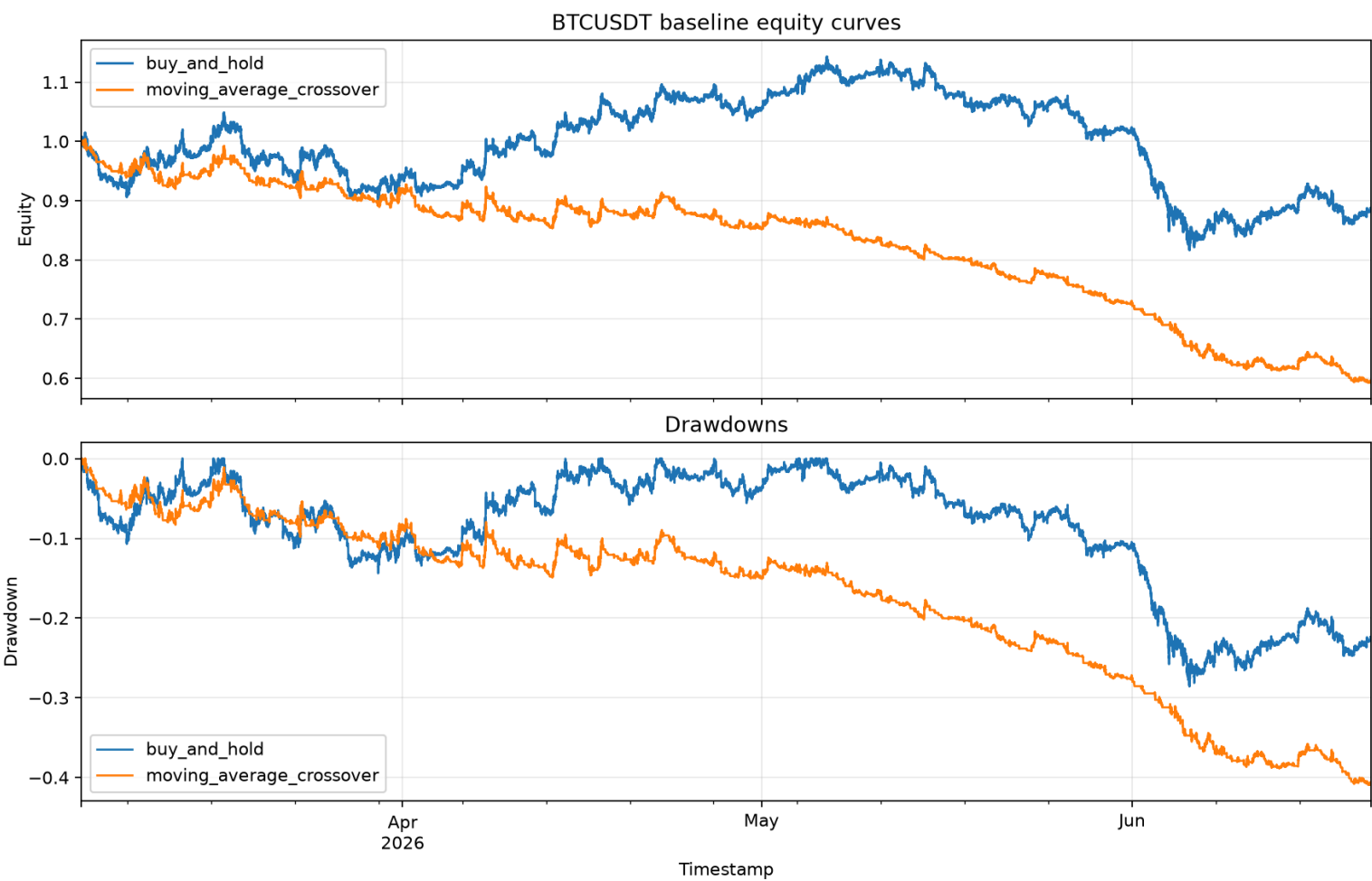
Data is not fetched implicitly during review: smoke runs use committed data; full runs use the documented external bundle.

3. Level 2.1: Single-Crypto Baseline

BTCUSDT moving-average crossover is the benchmark strategy

Strategy	Return	Sharpe	Max DD
Buy and hold	-11.26%	-0.72	-28.59%
MA crossover	-40.40%	-5.81	-40.91%

- Fast window: 60 minutes; slow window: 360 minutes.
- Transaction cost: 5 bps per turnover.
- Evaluation period: out-of-sample test only.
- Result: baseline is simple and interpretable, but not profitable after costs.

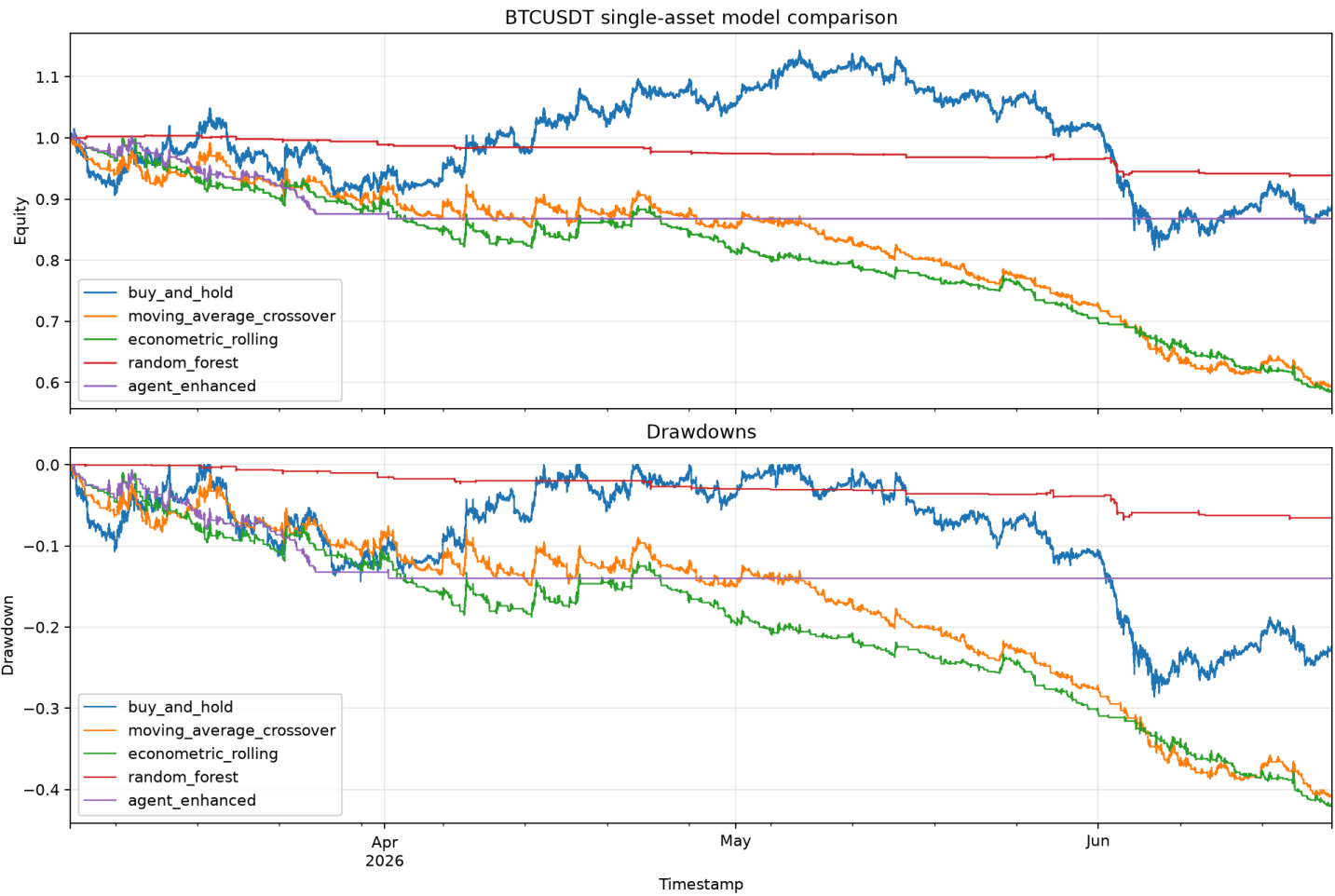


4. Level 2.2: Econometric, ML, and Agent Strategy

Single-pair model comparison uses one shared backtest layer

Strategy	Return	Sharpe	Max DD
Buy and hold	-11.26%	-0.72	-28.59%
MA crossover	-40.40%	-5.81	-40.91%
Econometric	-41.25%	-7.22	-42.07%
RandomForest	-6.13%	-4.89	-6.84%
Agent enhanced	-13.21%	-4.39	-13.99%

Features: lagged returns, rolling mean/volatility, momentum, MA distance.



5. Level 2.2: Targets, Training, and Robustness

The model target and validation design are explicitly documented

Target variable

Initial ML target: next-period return direction. Improvement experiment: future 60-minute return greater than a 10 bps cost buffer; neutral rows are dropped.

Training / testing

Models train only before the split timestamp. Validation-tuned models choose thresholds inside the train period and evaluate the final test period once.

Strategy	Return	Sharpe	Max DD
Buy and hold	−11.26%	−0.72	−28.59%
Cost-aware boosting	−6.00%	−2.99	−8.06%
Validation RF 60m	−15.24%	−5.47	−15.85%

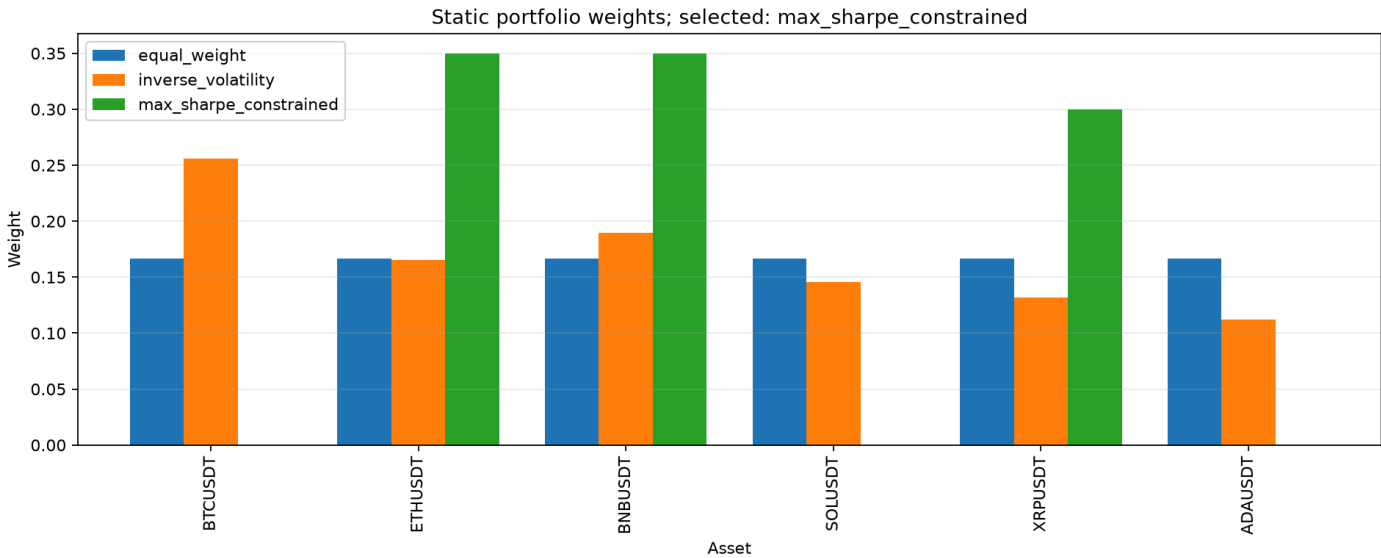
- Random chance check: out-of-sample-only scoring, validation threshold grid, and cost-aware labels.
- Retraining policy: periodic walk-forward refresh, not continuous online fitting in MVP.

6. Level 2.3: Static 6-Asset Portfolio

Historical portfolio management on BTC, ETH, BNB, SOL, XRP, ADA

Method	Return	Sharpe	Max DD
Equal weight	−19.41%	−1.25	−32.85%
Inverse vol	−17.28%	−1.13	−31.08%
Max-Sharpe	−14.97%	−0.96	−29.31%

- Selected method: constrained max-Sharpe by out-of-sample Sharpe.
- Selected weights: ETH 35%, BNB 35%, XRP 30%.
- Max asset weight cap: 35%.

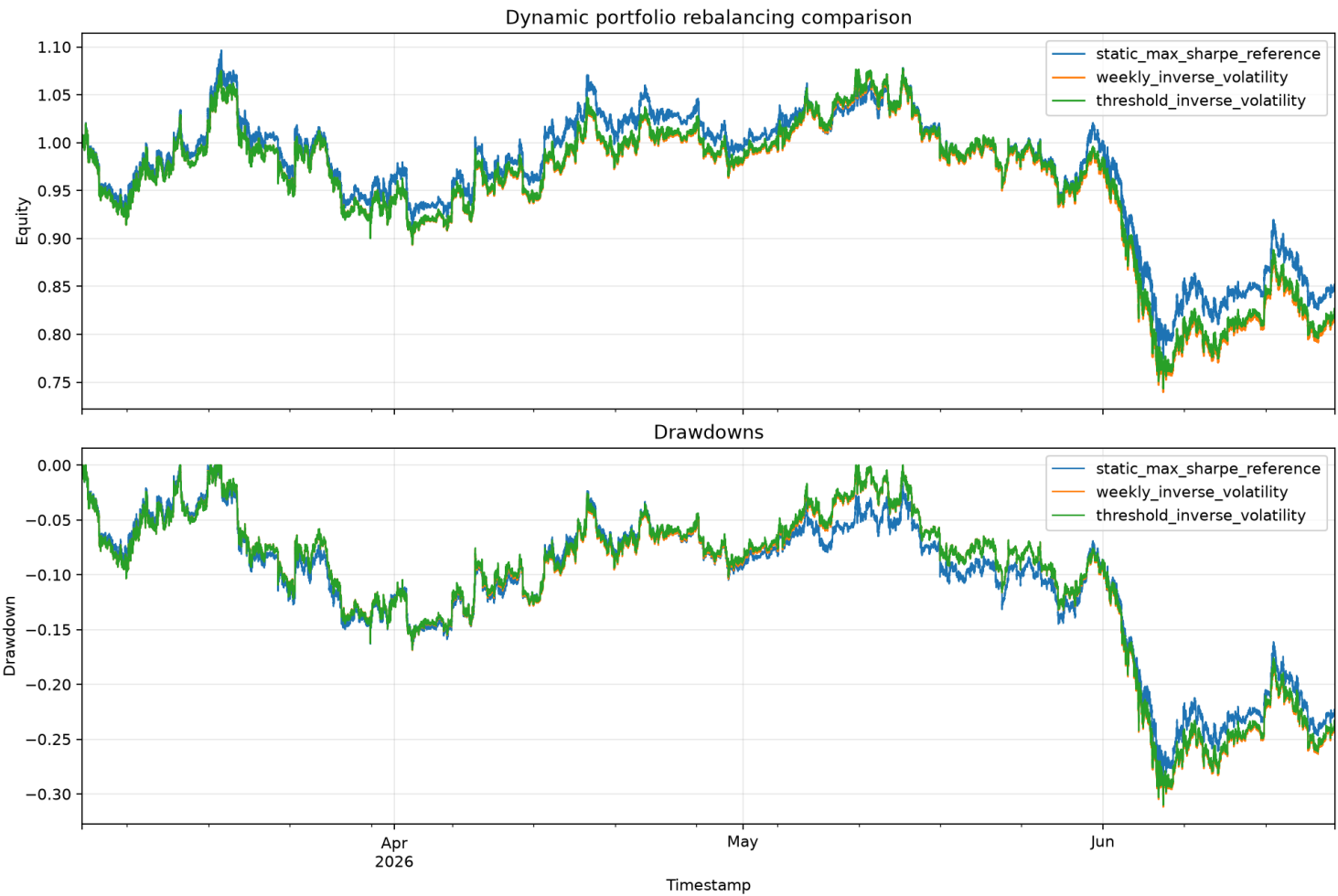


7. Level 2.4: Dynamic Rebalancing

Weekly and threshold-based policies are tested against the static reference

Strategy	Return	Sharpe	Events
Static reference	-14.97%	-0.96	0
Weekly inv-vol	-18.01%	-1.19	16
Threshold inv-vol	-17.55%	-1.14	5

- Logic: trailing inverse-volatility target weights.
- Threshold policy trades when passive drift crosses 2 percentage points.
- Selection criterion: highest out-of-sample Sharpe after costs.

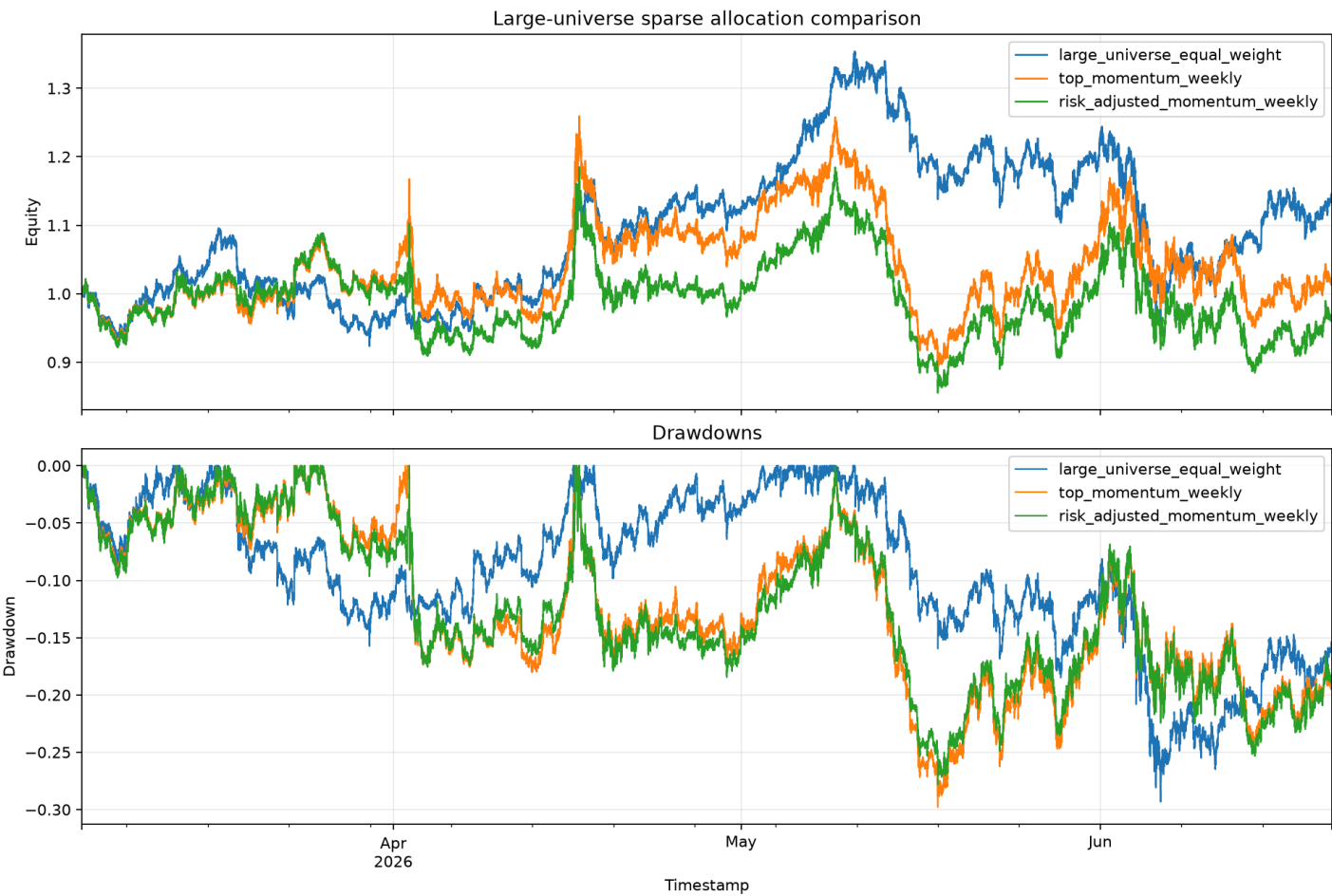


8. Level 2.5: Large-Universe Portfolio

The system scales to 120 USDT spot pairs

Strategy	Return	Sharpe	Max DD
120-pair equal weight	14.39%	1.18	−29.32%
Top momentum weekly	1.49%	0.41	−29.80%
Risk-adj. momentum	−3.78%	0.12	−27.83%

- Pair selection: coverage and liquidity first.
- Signal priority: top-20 weekly momentum or risk-adjusted momentum.
- Risk controls: active-count, max weight 8%, gross exposure floor.



9. Backtesting and Risk Metrics

Every strategy uses the same cost-aware evaluation contract

Backtest controls

- Chronological train/test split
- No shuffling for time series
- Signal lag prevents lookahead
- Turnover-based transaction costs
- Long-only exposure in MVP
- Same annualization convention across reports

Reported metrics

- Total and annualized return
- Annualized volatility
- Sharpe, Sortino, Calmar
- Max drawdown
- VaR and CVaR
- Hit rate, turnover, cost drag
- Effective assets and concentration

Why this matters

Comparing models without a shared cost and risk layer would overstate noisy high-frequency signals. The same evaluation layer makes all five assignment levels comparable.

10. Result Interpretation

The MVP is reproducible; the strongest current result is diversification

What worked

- RandomForest and cost-aware boosting reduced BTC drawdown versus buy-and-hold.
- Static constrained max-Sharpe improved the six-asset portfolio versus equal weight.
- 120-pair equal weight achieved positive out-of-sample return and Sharpe.
- Code, data manifest, reports, figures, and notebook are reproducible.

What did not work yet

- One-minute single-asset alpha remained noisy and cost-sensitive.
- Moving averages and rolling econometric signals overtraded.
- Dynamic rebalancing did not beat the static reference in the test window.
- Sparse momentum underperformed broad diversification in the large universe.

11. Reviewer Reproduction Map

Single notebook plus committed reports and data manifest

Primary artifacts

- Notebook: notebooks/final_solution.ipynb
- Package: src/ai_crypto_hedge_fund/
- Tests: tests/
- Metrics: reports/metrics/
- Figures: reports/figures/
- Data manifest: data/manifest.json

Validation commands

```
uv sync --locked
uv run pytest
DATA_MODE=sample uv run jupyter nbconvert \
  --to notebook --execute notebooks/final_solution.ipynb
```

Full 120-pair data is externally published and checksum-documented.

Conclusion: the submitted system satisfies the technical assignment as a reproducible research MVP; live execution remains future scope.